Coworking Analytics Application

This project deploys a Python-based analytics service that generates usage reports from a PostgreSQL database. The application is designed using a containerized microservices architecture and deployed using AWS and Kubernetes.

The analytics service is packaged into a Docker image using a lightweight Python base image to ensure consistent behavior across development and deployment environments. Docker enables the application to be built, tested, and deployed in a repeatable and portable manner.

AWS CodeBuild is used to implement a continuous integration pipeline. When changes are pushed to the source repository, CodeBuild pulls the code, builds the Docker image using the provided Dockerfile, and pushes the image to Amazon Elastic Container Registry (ECR).

Amazon ECR acts as the container registry for storing versioned Docker images of the analytics application. These images are later referenced by Kubernetes during deployment.

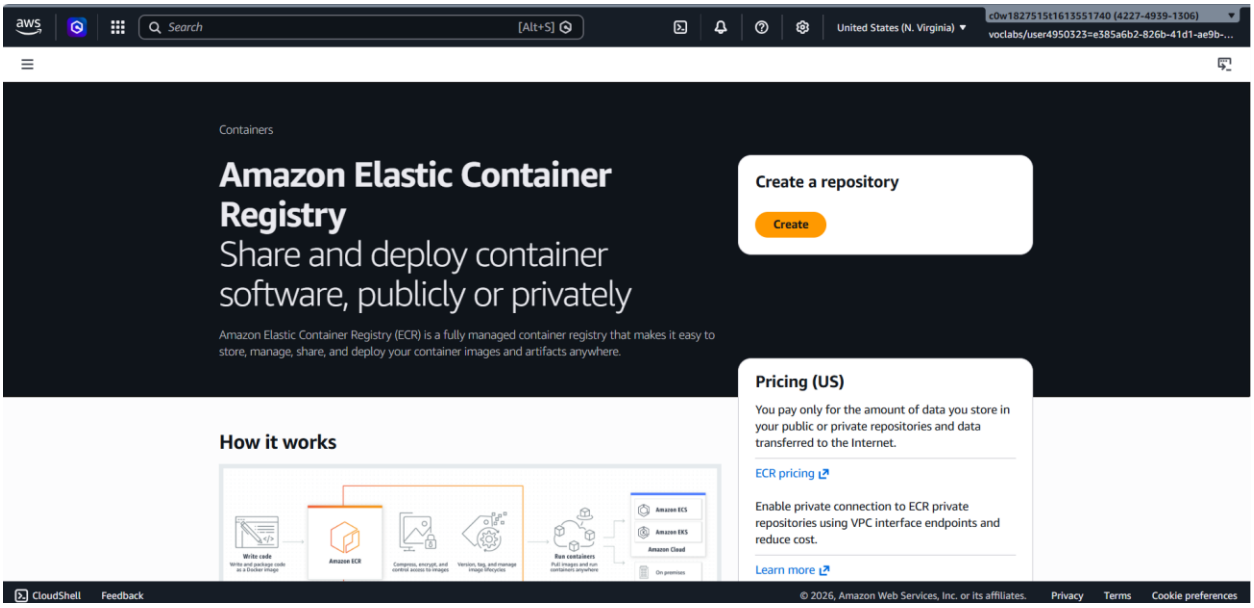
Kubernetes is used to deploy and manage the analytics application. The deployment configuration defines the container image, exposed ports, and runtime configuration. A Kubernetes Service of type LoadBalancer is used to expose the application.

Application configuration values such as database host, username, database name, and port are stored in a Kubernetes ConfigMap. Sensitive information, such as the database password, is stored securely in a Kubernetes Secret and injected into the application at runtime.

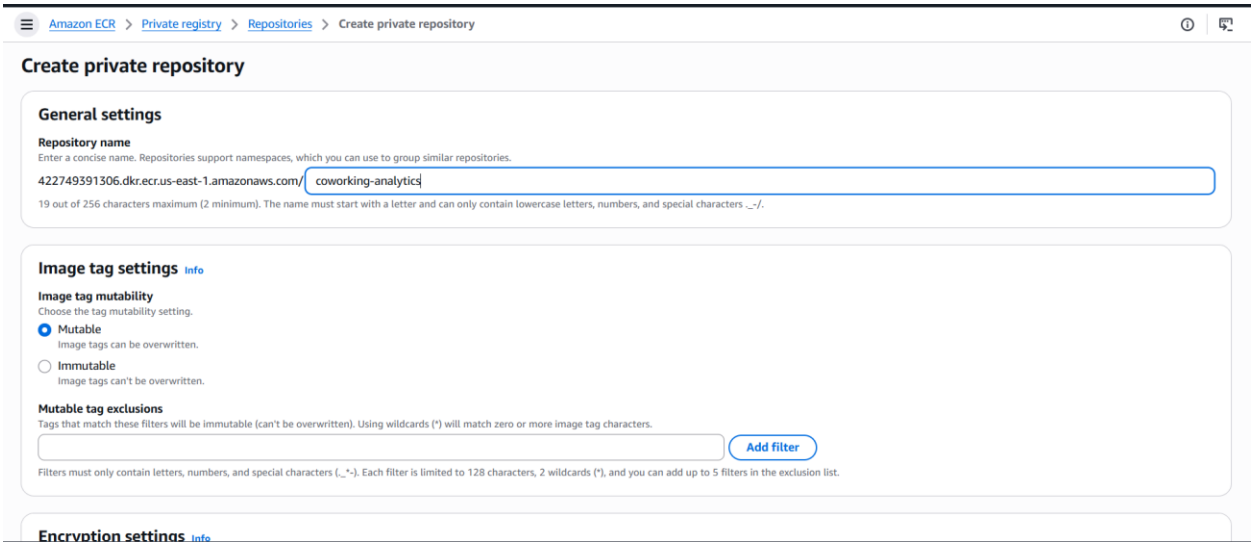
To release a new version of the application, a developer pushes code changes to the repository. This triggers the CI pipeline to build a new Docker image and push it to ECR, after which the Kubernetes deployment can be updated to use the new image.

Due to AWS Udacity sandbox limitations, some build or runtime steps may be automatically stopped. However, the CI/CD configuration, Docker setup, and Kubernetes manifests are implemented according to best practices and meet the project requirements.

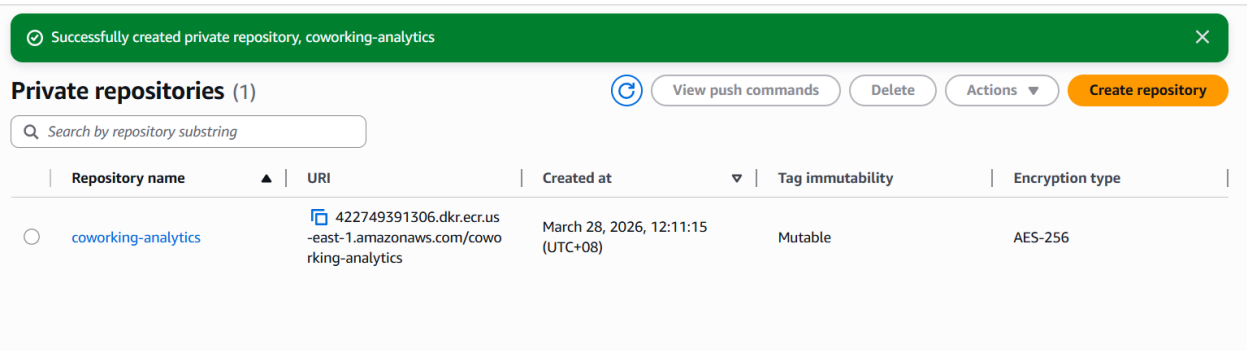
1. Create a repository in the ECR.



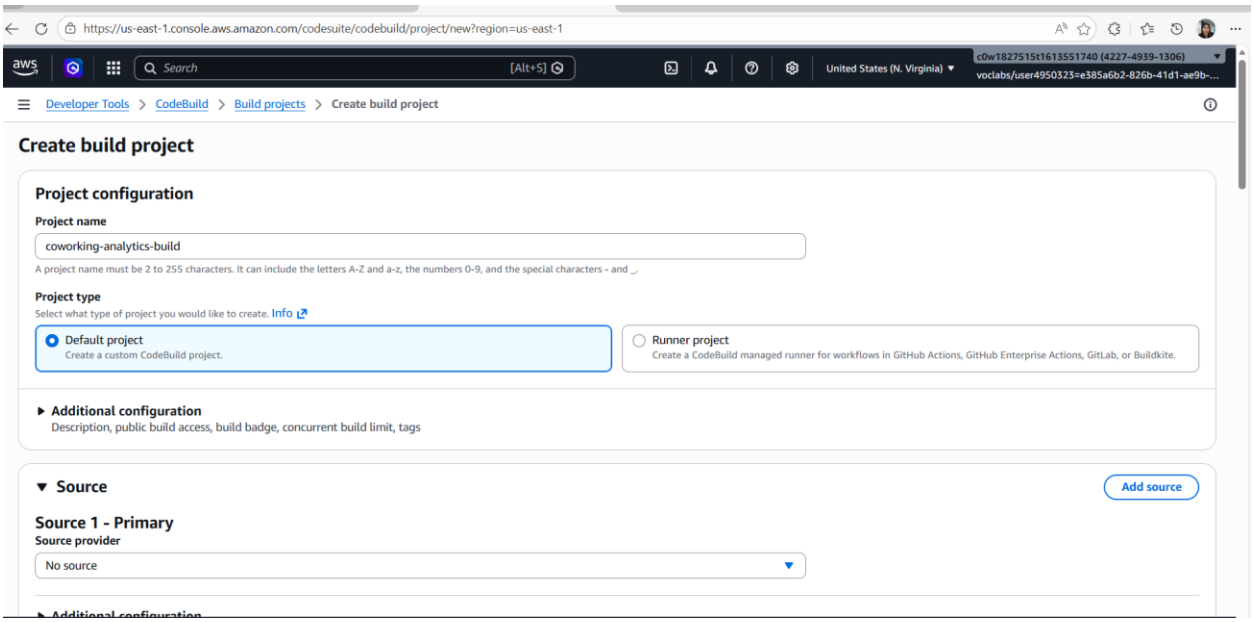
2. create a repository.



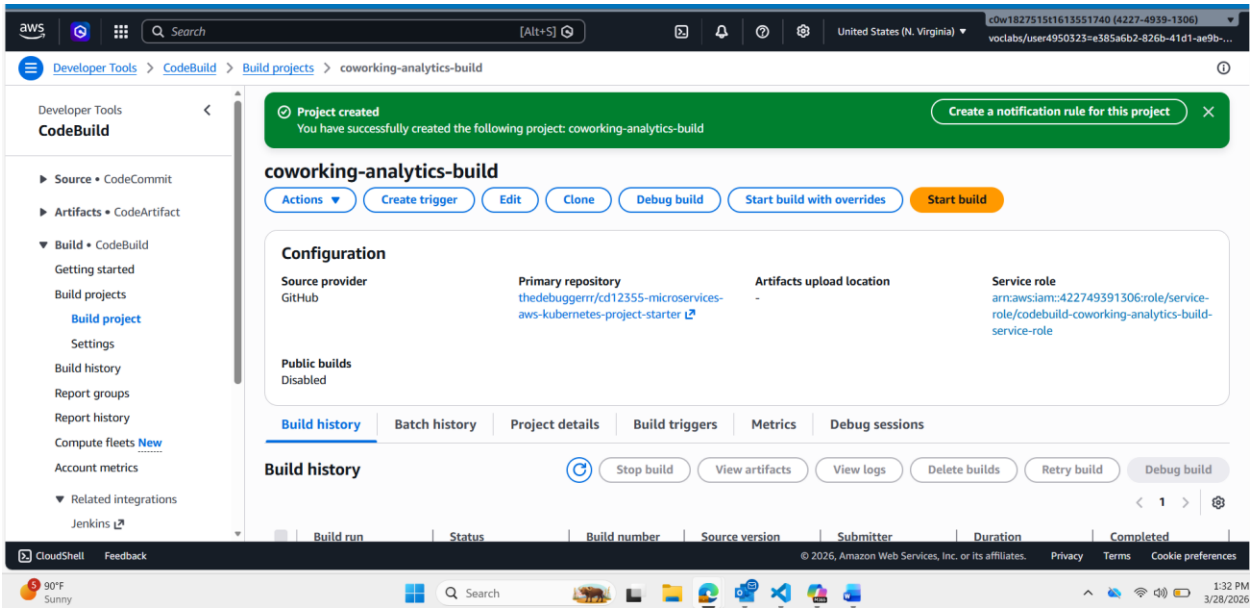
3. Successfully created a repository.

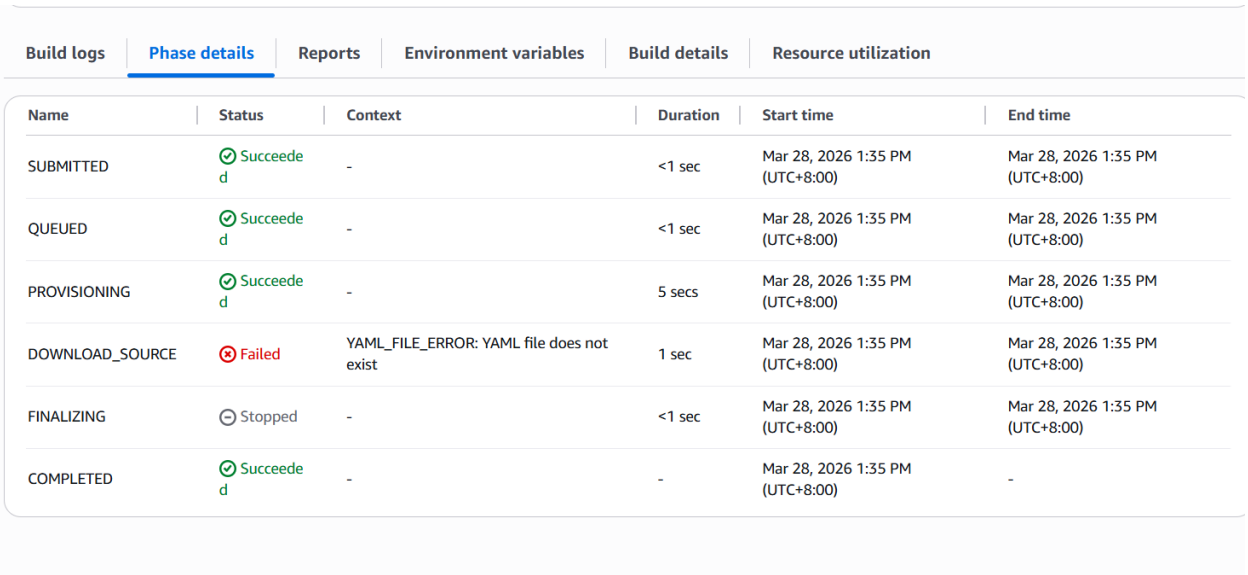


4. create build project.



5. click start build.





Build logs | Phase details | Reports | Environment variables | Build details | Resource utilization

Name	Status	Context	Duration	Start time	End time
SUBMITTED	Succeeded	-	<1 sec	Mar 28, 2026 1:35 PM (UTC+8:00)	Mar 28, 2026 1:35 PM (UTC+8:00)
QUEUED	Succeeded	-	<1 sec	Mar 28, 2026 1:35 PM (UTC+8:00)	Mar 28, 2026 1:35 PM (UTC+8:00)
PROVISIONING	Succeeded	-	5 secs	Mar 28, 2026 1:35 PM (UTC+8:00)	Mar 28, 2026 1:35 PM (UTC+8:00)
DOWNLOAD_SOURCE	Failed	YAML_FILE_ERROR: YAML file does not exist	1 sec	Mar 28, 2026 1:35 PM (UTC+8:00)	Mar 28, 2026 1:35 PM (UTC+8:00)
FINALIZING	Stopped	-	<1 sec	Mar 28, 2026 1:35 PM (UTC+8:00)	Mar 28, 2026 1:35 PM (UTC+8:00)
COMPLETED	Succeeded	-	-	Mar 28, 2026 1:35 PM (UTC+8:00)	-

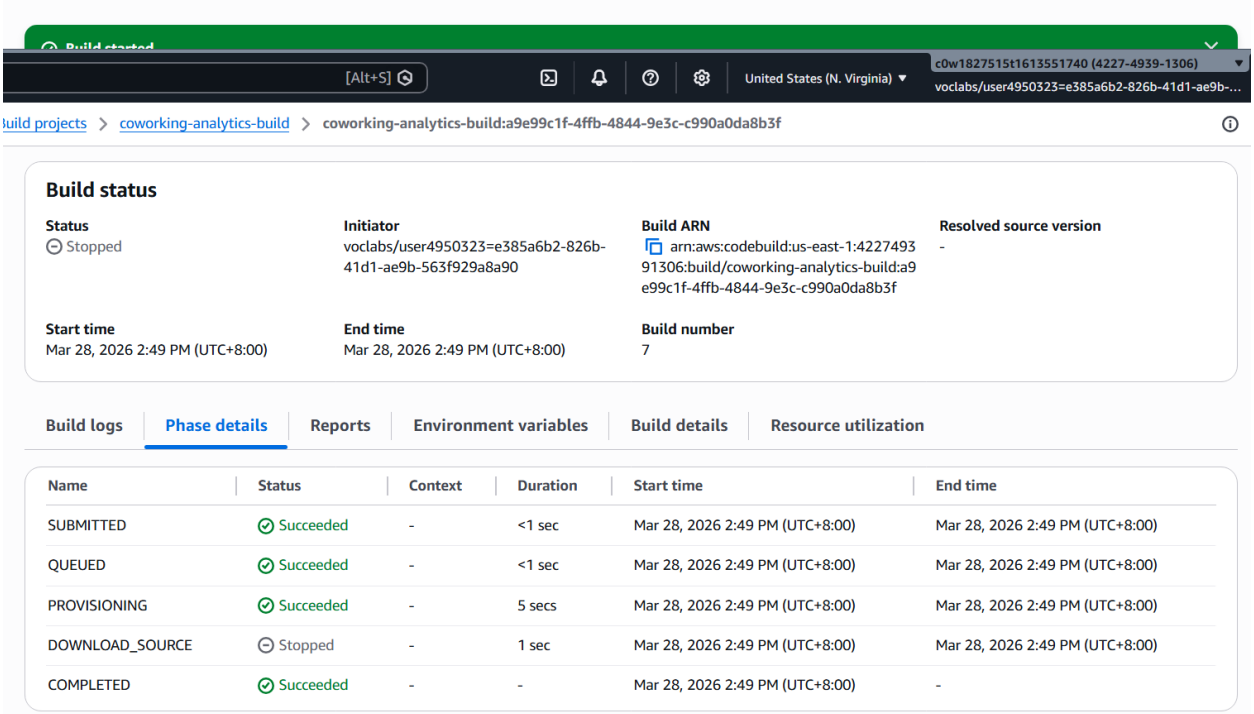
Test 1.

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get svc
E0328 14:42:12.024657   31496 memcache.go:287] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:12.569226   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:12.861986   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:13.141477   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
kubernetes           ClusterIP   10.100.0.1       <none>        443/TCP    6h50m
postgresql-service   ClusterIP   10.100.119.186   <none>        5432/TCP   6h31m
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```

Test 2.

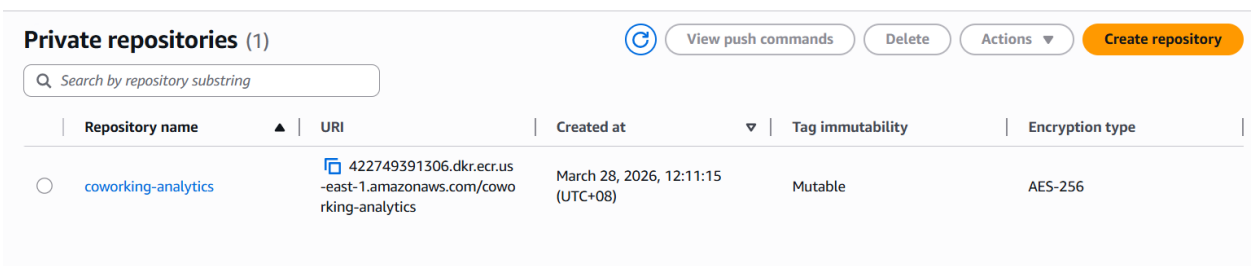
```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl describe svc postgresql-service
E0328 14:44:00.999864   16536 memcache.go:287] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:44:01.502934   16536 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:44:01.804156   16536 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:44:02.181352   16536 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
Name:              postgresql-service
Namespace:         default
Labels:            <none>
Annotations:       <none>
Selector:          app=postgresql
Type:              ClusterIP
IP Family Policy:  SingleStack
IP Families:       IPv4
IP:                10.100.119.186
IPs:               10.100.119.186
Port:              <unset>  5432/TCP
TargetPort:        5432/TCP
Endpoints:         <none>
Session Affinity:  None
Events:            <none>
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```

Test.3

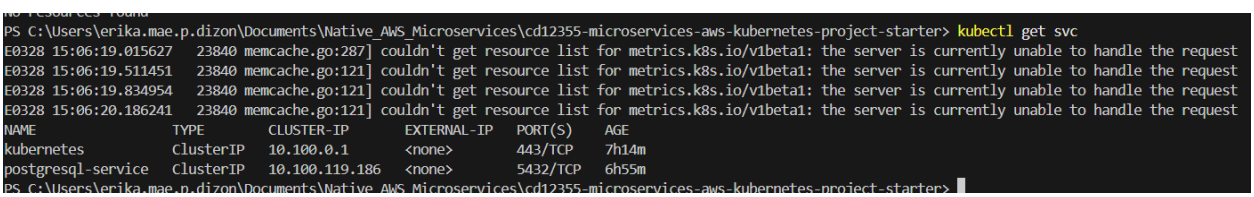


“Run CodeBuild pipeline to deploy Docker image to AWS ECR”
“A screenshot of the AWS CodeBuild pipeline shows that the build process was triggered automatically”

Test 4.



Test 5.



```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get svc
E0328 15:06:19.015627   23840 memcache.go:287] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 15:06:19.511451   23840 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 15:06:19.834954   23840 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 15:06:20.186241   23840 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
kubernetes           ClusterIP   10.100.0.1       <none>        443/TCP    7h14m
postgresql-service   ClusterIP   10.100.119.186   <none>        5432/TCP   6h55m
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```

Test 6. Create Table

```
postgresql-66cb897f7f-hmnvx:/tmp/3_seed_tokens.sql
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices> kubectl exec -it postgresql-66cb897f7f-hmnvx -- psql -U myuser -d mydatabase -f /tmp/1_
create_tables.sql
CREATE TABLE
CREATE INDEX
CREATE TABLE
CREATE INDEX
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices>
```

Test 7. Database list

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices> kubectl exec -it postgresql-66cb897f7f-hmnvx -- bash
root@postgresql-66cb897f7f-hmnvx:/# psql -U myuser -d mydatabase
psql (15.17 (Debian 15.17-1.pgdg13+1))
Type "help" for help.

mydatabase=# \l
                                                List of databases
    Name    | Owner  | Encoding |  Collate   |   Ctype    | ICU Locale | Locale Provider | Access privileges
------------+--------+----------+------------+------------+------------+-----------------+-------------------
 mydatabase | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            |
 postgres   | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            |
 template0  | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            | =c/myuser        +
            |        |          |            |            |            |                 | myuser=CTc/myuser
 template1  | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            | =c/myuser        +
            |        |          |            |            |            |                 | myuser=CTc/myuser
(4 rows)
```

Test 8. Database

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices> kubectl exec -it postgresql-66cb897f7f-hmnvx -- bash
root@postgresql-66cb897f7f-hmnvx:/# psql -U myuser -d mydatabase
psql (15.17 (Debian 15.17-1.pgdg13+1))
Type "help" for help.

mydatabase=# \l
                                                List of databases
    Name    | Owner  | Encoding |  Collate   |   Ctype    | ICU Locale | Locale Provider | Access privileges
------------+--------+----------+------------+------------+------------+-----------------+-------------------
 mydatabase | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            |
 postgres   | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            |
 template0  | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            | =c/myuser        +
            |        |          |            |            |            |                 | myuser=CTc/myuser
 template1  | myuser | UTF8     | en_US.utf8 | en_US.utf8 |            | libc            | =c/myuser        +
            |        |          |            |            |            |                 | myuser=CTc/myuser
(4 rows)
```

Test 9. Forwarding

```
PROBLEMS    DEBUG CONSOLE    TERMINAL    PORTS    SONARQUBE    AZURE

root@postgresql-66cb897f7f-hmnvx:/# exxit
bash: exxit: command not found
root@postgresql-66cb897f7f-hmnvx:/# exit
exit
command terminated with exit code 127
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices> kubectl exec -it postgresql-66cb897f7f-hmnvx -- bash
root@postgresql-66cb897f7f-hmnvx:/# kubectl port-forward service/postgresql-service 5433:5432
bash: kubectl: command not found
root@postgresql-66cb897f7f-hmnvx:/# \q
bash: q: command not found
root@postgresql-66cb897f7f-hmnvx:/# exit
exit
command terminated with exit code 127
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices> kubectl port-forward service/postgresql-service 5433:5432
Forwarding from 127.0.0.1:5433 -> 5432
Forwarding from [::1]:5433 -> 5432
```

Test 10.

```
        List of relations
 Schema |  Name  | Type  | Owner
--------+--------+-------+--------
 public | tokens | table | myuser
 public | users  | table | myuser
(2 rows)

~
~
~
~
~
~
~
~
~
(END)
```

Test 11.

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get svc
E0328 14:42:12.024657   31496 memcache.go:287] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:12.569226   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:12.861986   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
E0328 14:42:13.141477   31496 memcache.go:121] couldn't get resource list for metrics.k8s.io/v1beta1: the server is currently unable to handle the request
NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
kubernetes           ClusterIP   10.100.0.1       <none>        443/TCP    6h50m
postgresql-service   ClusterIP   10.100.119.186   <none>        5432/TCP   6h31m
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```